

Surface Pattern Recognition Using Shape Subspace

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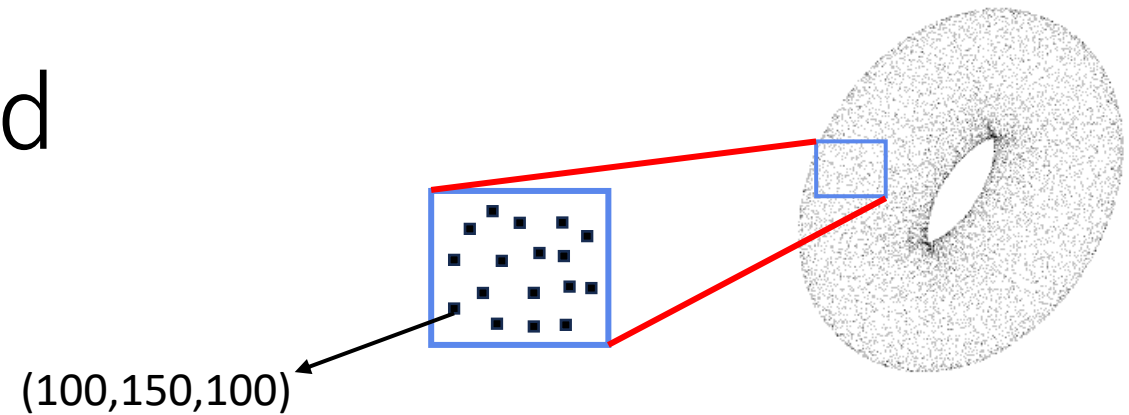
Research Problem

- How can Subspace-based methods be used to detect surface irregularities in Point Cloud data?



Research Background

Point Cloud Processing



- Point Cloud processing involves handling 3D data represented as a collection of points in space (x, y, z) coordinates.
- Example of point cloud data file

X	Y	Z
100	150	100
200	200	210
10	50	100
35	10	35

Coordinates

R	G	B
13	115	24
32	102	35
15	57	78
33	43	25

Color (Optional)

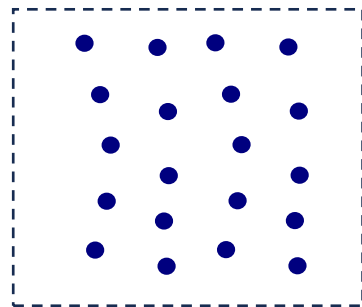
X	Y	Z
1	0	3
-1	1	0
0	-2	0
2	-1	1

Normal Vectors (Optional)

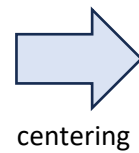
Research Background

Shape Subspace

- A mathematical representation of 3D structures based on the relative positions of feature points.
 - Used for task such as object recognition, motion analysis, and biometric authentication.
 - Representing high-dimensional data efficiently.



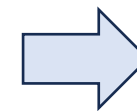
Input point cloud



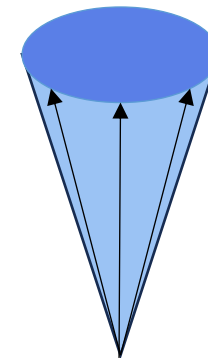
centering

$$\mathbf{V}_{\text{centered}} = \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} & z_1 - \bar{z} \\ x_2 - \bar{x} & y_2 - \bar{y} & z_2 - \bar{z} \\ \vdots & \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} & z_n - \bar{z} \end{bmatrix}$$

Centered matrix



Apply SVD



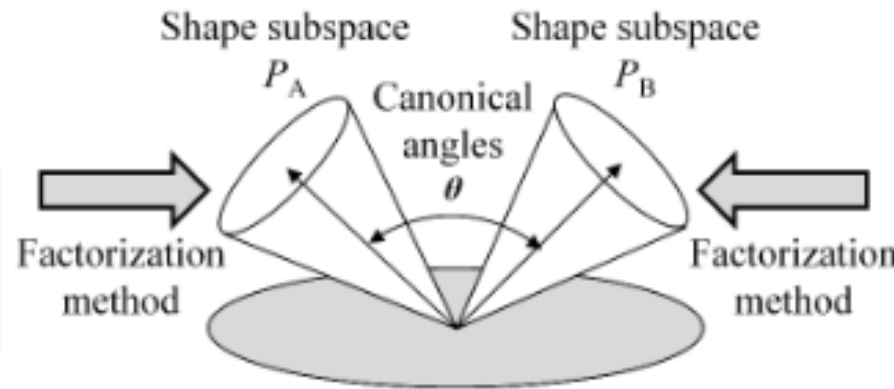
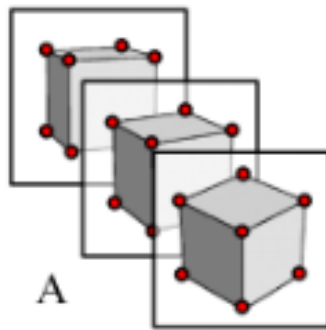
Shape Subspace

Related Research

How is Shape Subspace computed?

Factorization Approach

Frames of an Image sequence



Frames of an Image sequence

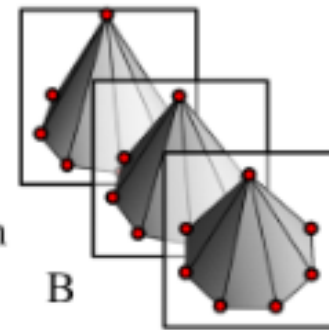


Figure adapted from Igarashi and Fukui, "3D Object Recognition Based on Canonical Angles between Shape Subspaces"

Related Research

How is Shape Subspace computed?

Direct 3D Feature Extraction Approach

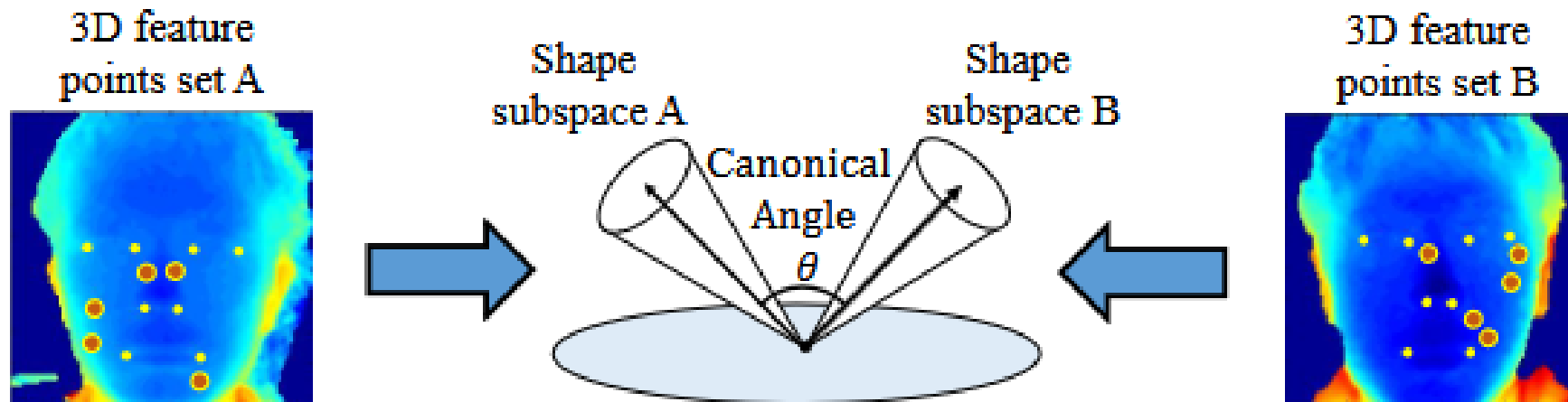


Figure adapted from Yoshinuma, Hino, and Fukui, "Personal Authentication Based on 3D Configuration of Micro-feature Points on Facial Surface"

Related Research

- The Difference Subspace (DS) is a geometric concept that represents the components of difference between two class subspaces. It extends the idea of a difference vector between two vectors to the case of subspaces.
- DS is particularly useful for analyzing and extracting local shape differences between objects, which provides a more sensitive cue for recognition compared to traditional global shape comparison methods.

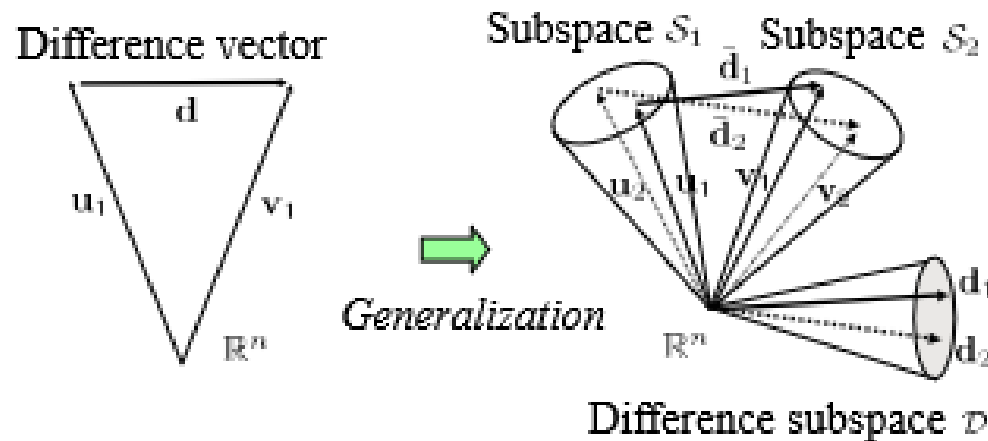
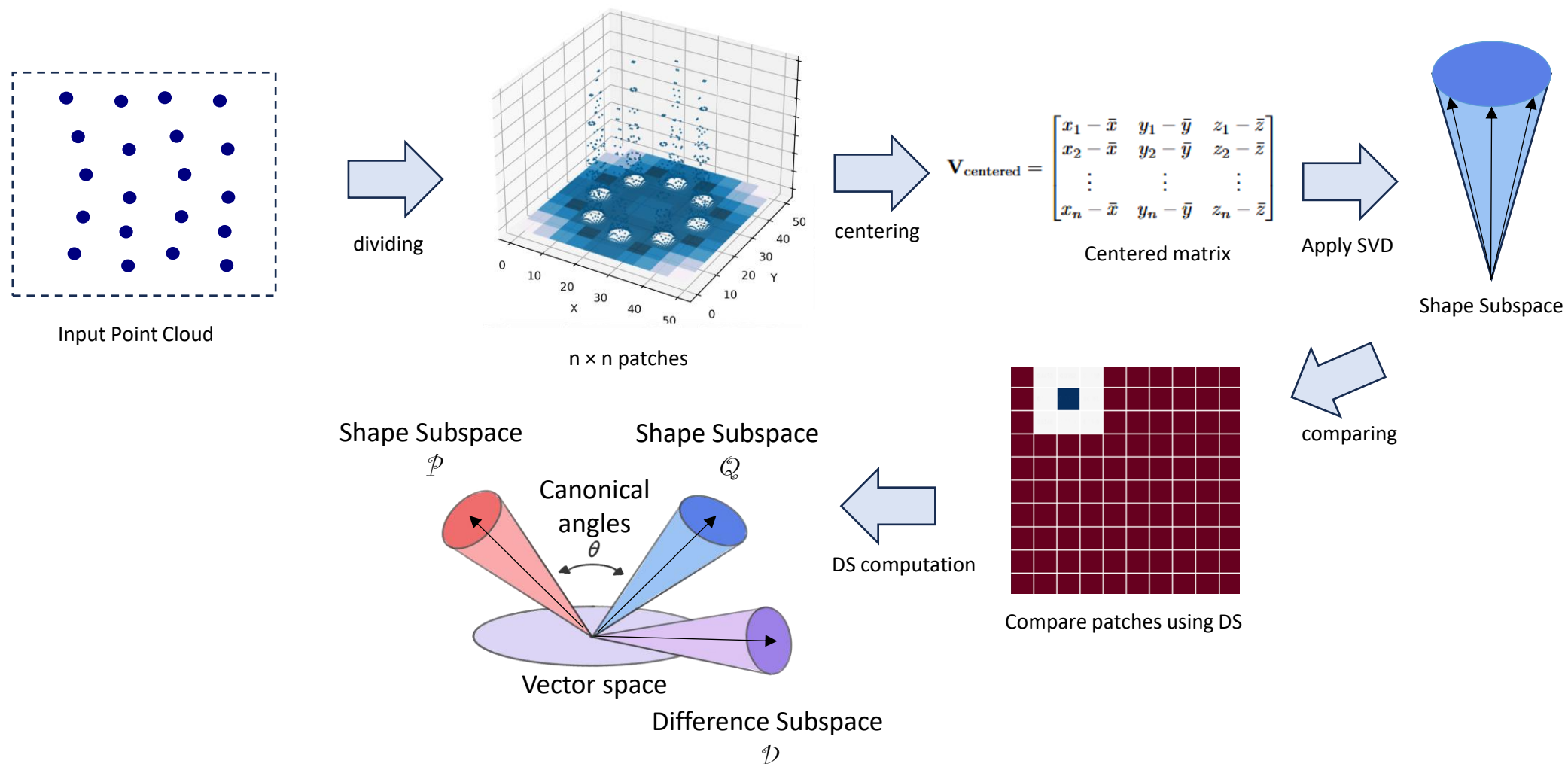


Figure adapted from Fukui et al., "Second-order Difference Subspace"

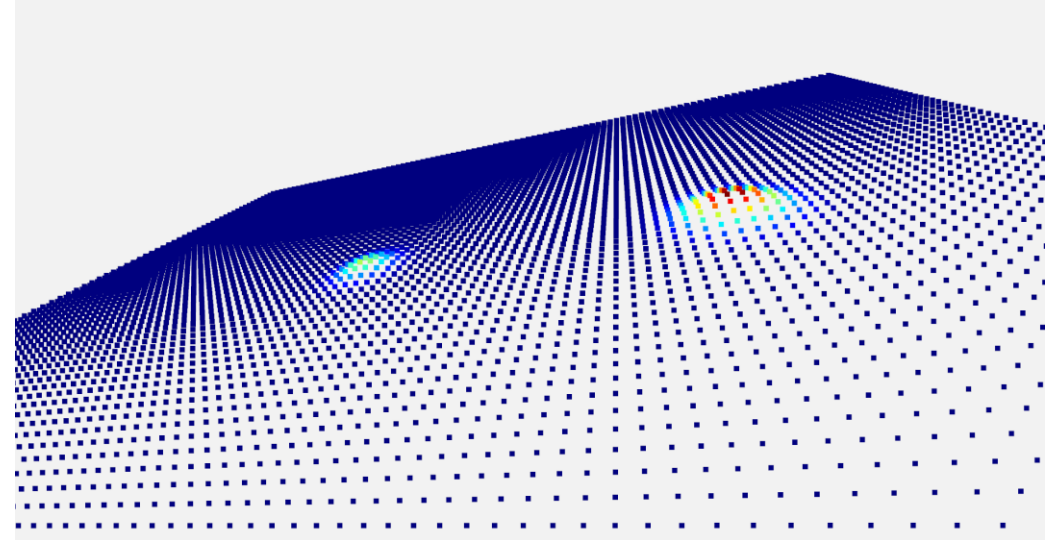
Proposed Method



Experiment

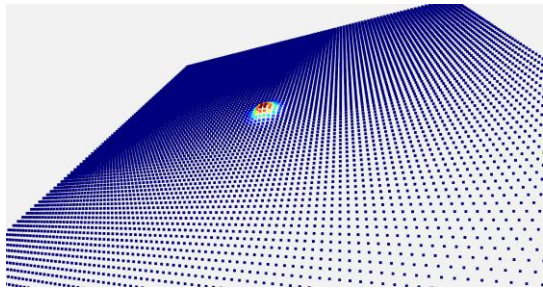
1. Generation of synthetic floor.

- Configuration:
 - The synthetic floor is defined by dimensions of 50×50 where each point is spaced exactly one centimeter apart
 - Total points 10,000

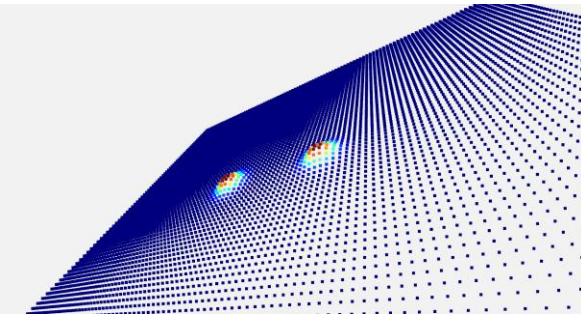


Experiment

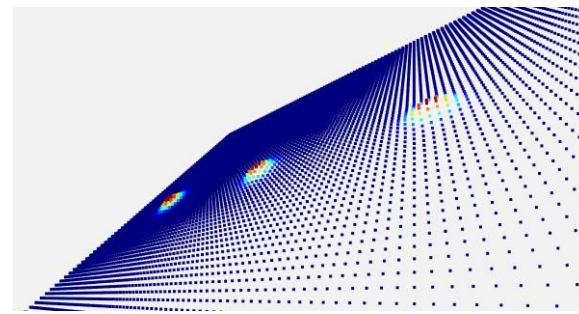
2. Generating predefined or random bumps



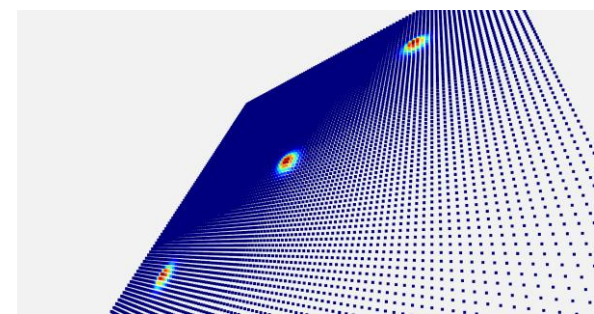
Single



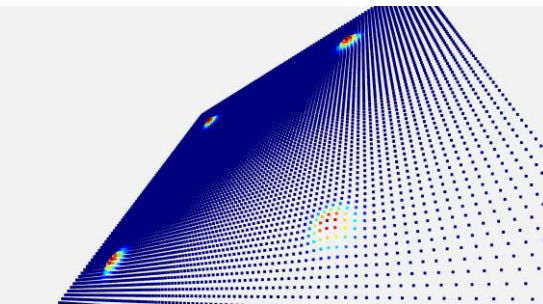
Double



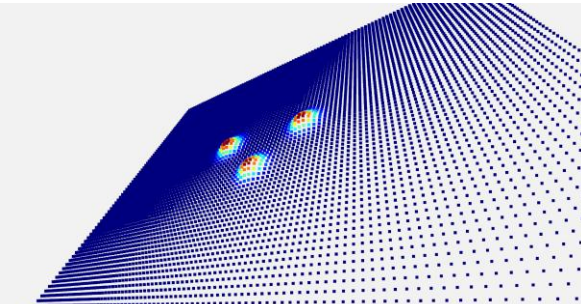
Ridge



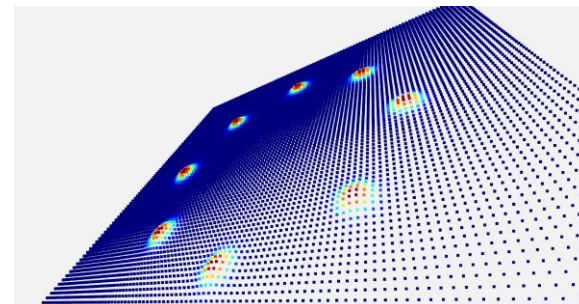
Diagonal



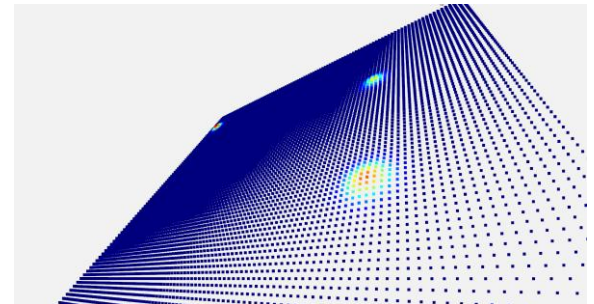
Corners



Triangle



Circle

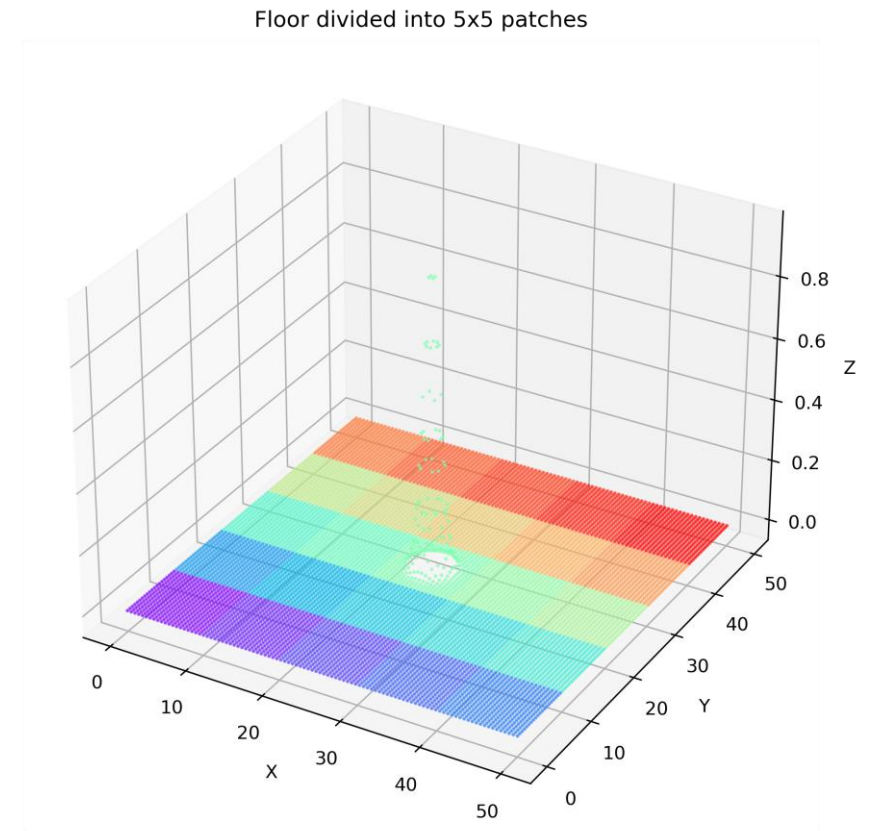


Random

Experiment

2. Patch division process

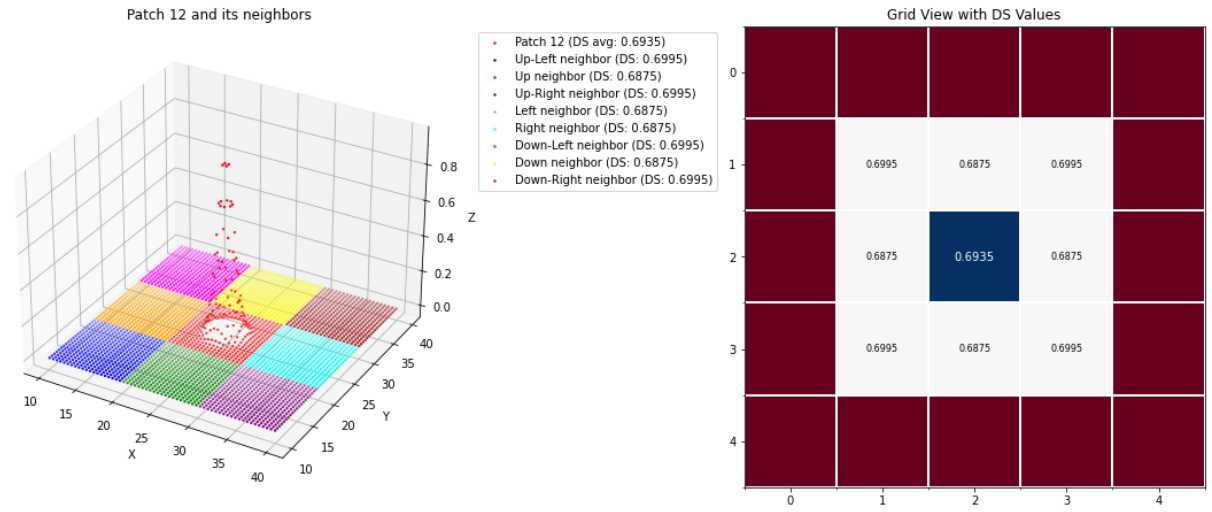
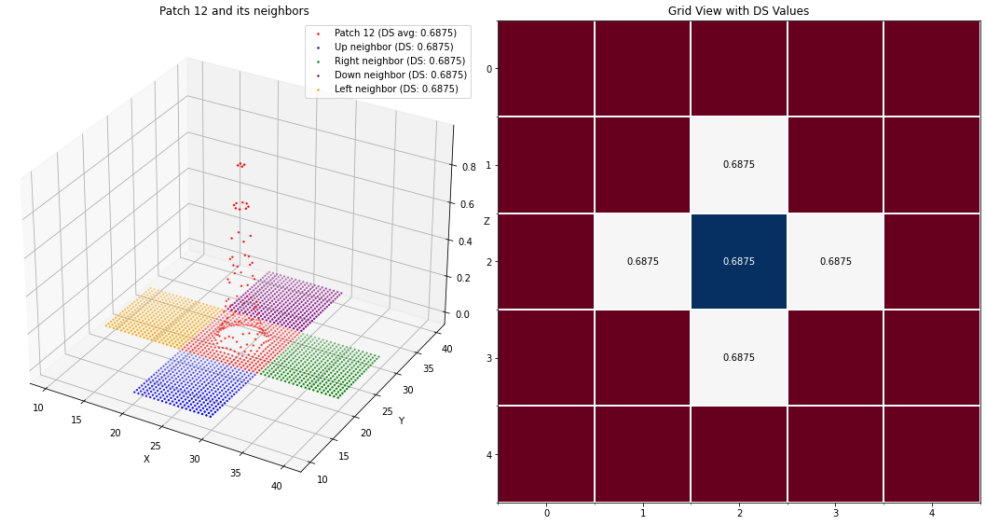
- Configuration:
 - Number of patches along x-axis
 - Number of patches along y-axis



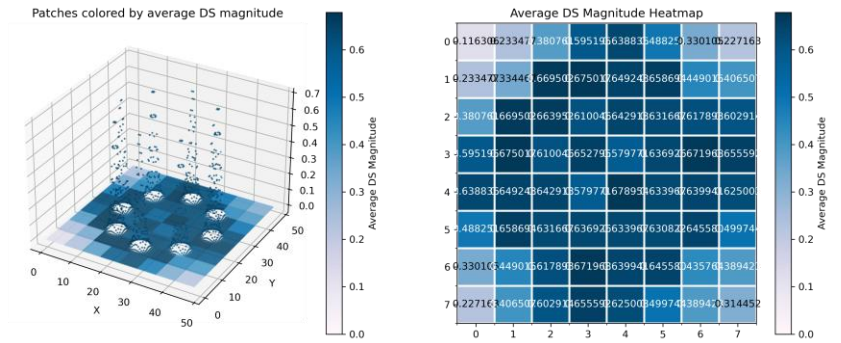
Experiment

3. DS generation for each patch

- 4-neighbor approach
- 8-neighbor approach



Summary of Initial Research



- **Methodology**

- **Data:** Synthetic 3D floors (50×50 units, 10,000 points) allowed controlled testing with known anomalies.
- **Segmentation:** Floors divided into uniform, non-overlapping patches (e.g., 8×8).
- **Shape Subspace:** Computed using SVD on each patch's centered matrix.
- **DS Calculation:** Calculated between neighboring patches to quantify local geometric differences.

- **Key Results**

- **Pattern Detection:** Successfully detected synthetic patterns (e.g., bumps, ridges, corners, circles, triangles, random).
- **Accuracy & Stability:**
 - Smaller patches (3×3, 4×4) improved fine-detail detection.
 - Eight-neighbor comparisons enhanced stability for complex patterns.

Limitations Identified in Previous Work

Boundary Sensitivity

- *Issue*: Non-overlapping patches may cut across anomalies, weakening signals near edges.
- *Impact*: Less accurate localization; higher risk of false negatives.

Limited Spatial Resolution

- **Unvalidated on Real-***Issue*: Sparse, discrete patches limited surface detail.
- *Impact*: Coarse results; harder to detect and locate fine anomalies.

World Data

- *Issue*: Tested only on clean synthetic data.
- *Impact*: Real-world robustness under noise and irregularities was unproven.

Sliding Window Approach



- **Concept**

- **Overlapping Analysis Regions:** Instead of fixed, discrete patches, the sliding window introduces analysis regions that overlap with configurable percentages.
- **Continuous Coverage:** This overlap ensures that every point on the surface is analyzed multiple times from different window perspectives, mitigating the boundary sensitivity issues of fixed patches.
- **Increased Spatial Resolution:** By generating more analysis points (windows) across the surface, the method achieves a much finer spatial resolution for anomaly localization and characterization.

Methodology

- Given a point cloud with overall bounding box dimensions (W,H) in the horizontal plane:
 - **Window Size:** (w,h) represents the dimensions of each individual analysis window.
 - **Overlap Percentage:** $\alpha \in [0,1)$ defines the degree of overlap between adjacent windows (e.g., $\alpha=0.5$ for 50% overlap).
- The **stride parameters** (sx,sy) — the distance the window moves for each step in X and Y directions — are calculated as:

$$s_x = w \cdot (1 - \alpha)$$

$$s_y = h \cdot (1 - \alpha)$$

Methodology

- The resulting **grid dimensions** (N_x, N_y) — the total number of windows in the X and Y directions become:

$$N_x = \left\lfloor \frac{s_x}{W - w} \right\rfloor + 1$$

$$N_y = \left\lfloor \frac{s_y}{H - h} \right\rfloor + 1$$

- Each window $W_{i,j}$ at grid position (i,j) encompasses points within the region defined by:

$$R_{i,j} = \{(x, y, z) \mid x_{\min} + i \cdot s_x \leq x < x_{\min} + i \cdot s_x + w\}$$

$$R_{i,j} = \{(x, y, z) \mid x_{\min} + i \cdot s_x \leq x < x_{\min} + i \cdot s_x + w, y_{\min} + j \cdot s_y \leq y < y_{\min} + j \cdot s_y + h\}$$

- where $(x_{\min}, y_{\min})$ are the minimum coordinates of the overall point cloud.

Experimental Validation

- **Comparison Methods**

- **Fixed Patches (Baseline):** 8×8 grid (64 patches), used as reference.
- **Sliding Window (Proposed):** 50% overlap between windows, increasing analysis density.

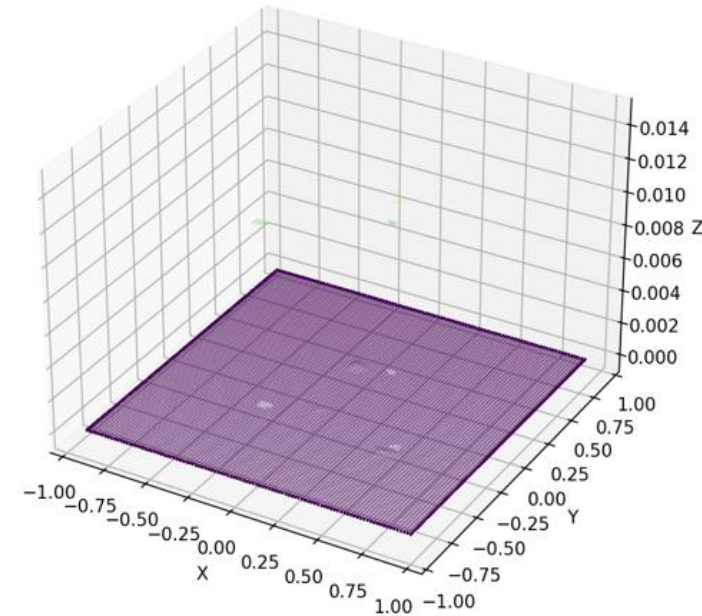
- **Data Used**

- **Synthetic Data:**

- Artificial 3D floor (50×50 units, ~10,000 points)
- Includes predefined patterns (bumps, ridges, corners, etc.)

- **Real-World Data:**

- Format: PLY (Polygon File)
- 61,914 points from a noisy floor scan



Experimental Validation

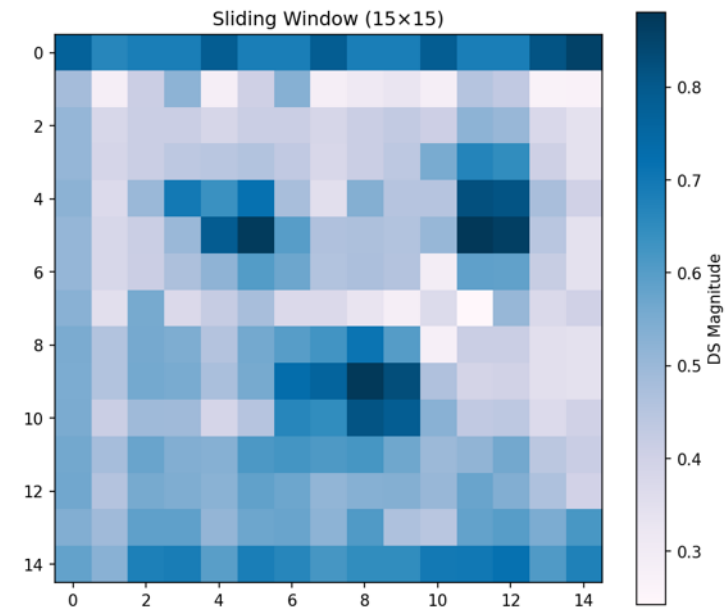
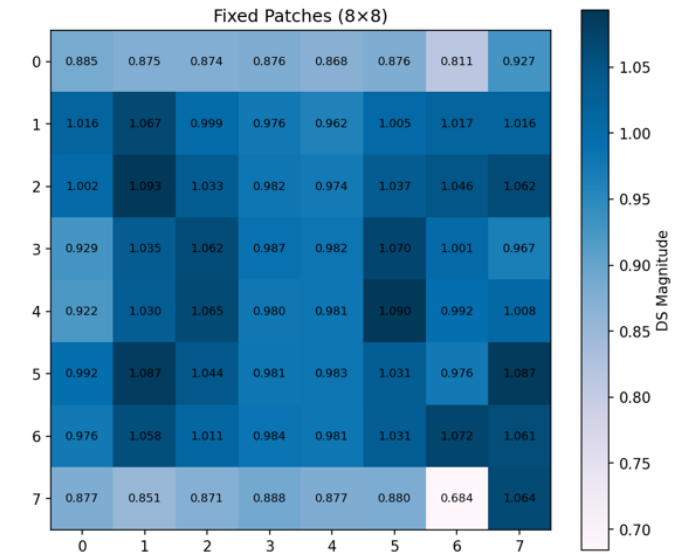
- **Common Settings**

- **Subspace Dim:** Set to 3 for all computations.
- **Neighbors:** Used 8-neighbor DS calculation for context.
- **Sliding Overlap:** Fixed at 50% for better coverage and resolution.

- **Preprocessing**

- **Centering:** Removed translation for geometry-only focus.
- **Downsampling (Optional):** Reduced density to ease computation.
- **Min Point Filter:** Ensured enough points per patch/window for stable SVD.

PLY File Analysis: Fixed Patches vs Sliding Window



Connecting Previous Work

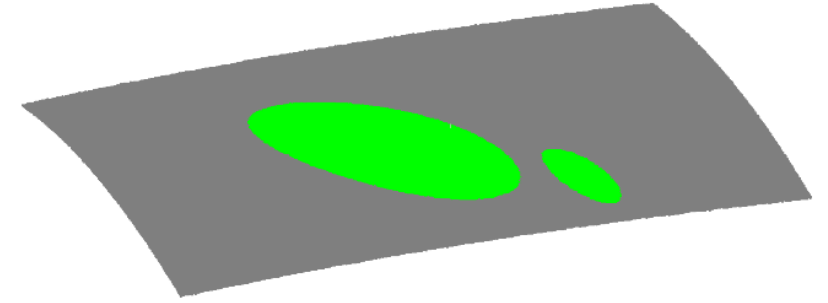
Recap of Initial Research

We explored Shape Subspace methods on synthetic floors.

- **Fixed Patch Analysis:** Successfully detected known patterns (bumps, ridges).
- **Sliding Window Extension:** Provided 3.5x higher resolution and handled noisy, real-world PLY files more effectively.

A Solution: DentNet's 2.5D Strategy

Handling 3D Data Efficiently



The DentNet paper introduces an elegant 2.5D strategy:

- **Quadric Surface Fitting:** A bivariate quadratic function, $z(x,y)$, is fit to the 3D point cloud to capture the baseline aircraft curvature.
- **Residual Calculation:** The distance of each point from this fitted surface is calculated. Dents appear as high-residual regions.

This effectively converts the complex 3D data into a 2D residual map, which is fast, memory-efficient, and highlights deformations for a CNN.

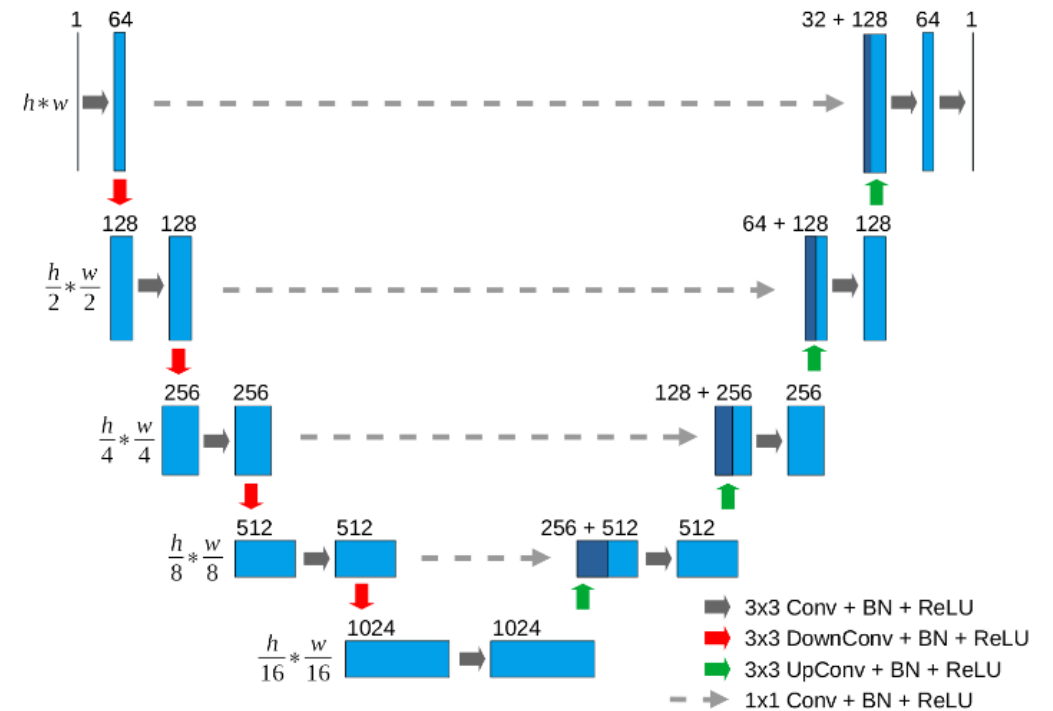
DentNet Architecture (U-Net FCN)

A U-Net for Segmentation

DentNet uses a Fully Convolutional Network based on the U-Net architecture.

- **Encoder (Downsampling):** Extracts features at multiple scales, with a 1024-channel bottleneck.
- **Decoder (Upsampling):** Reconstructs the full-size segmentation mask.
- **Skip Connections:** Propagate fine details from the encoder to the decoder, crucial for precise boundaries.

The model was trained on 30,000 synthetic samples and achieved **99.24% IoU** (Intersection over Union) on synthetic validation data.



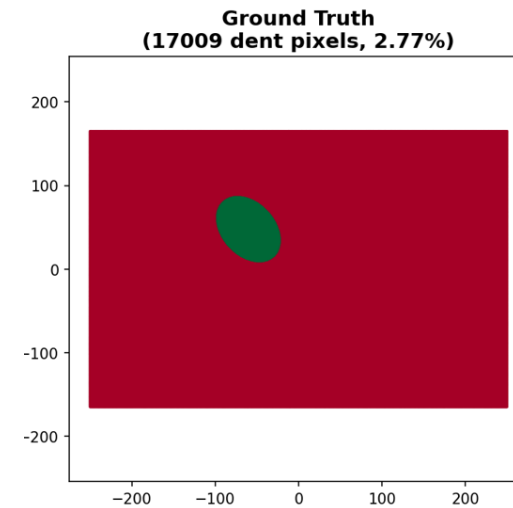
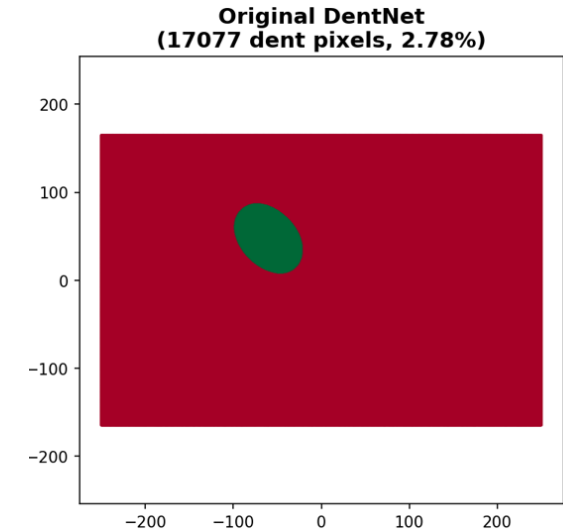
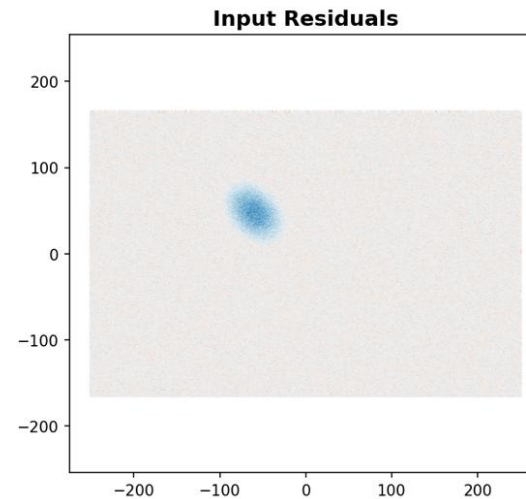
DentNet Synthetic Test Results

Performance on Synthetic Data

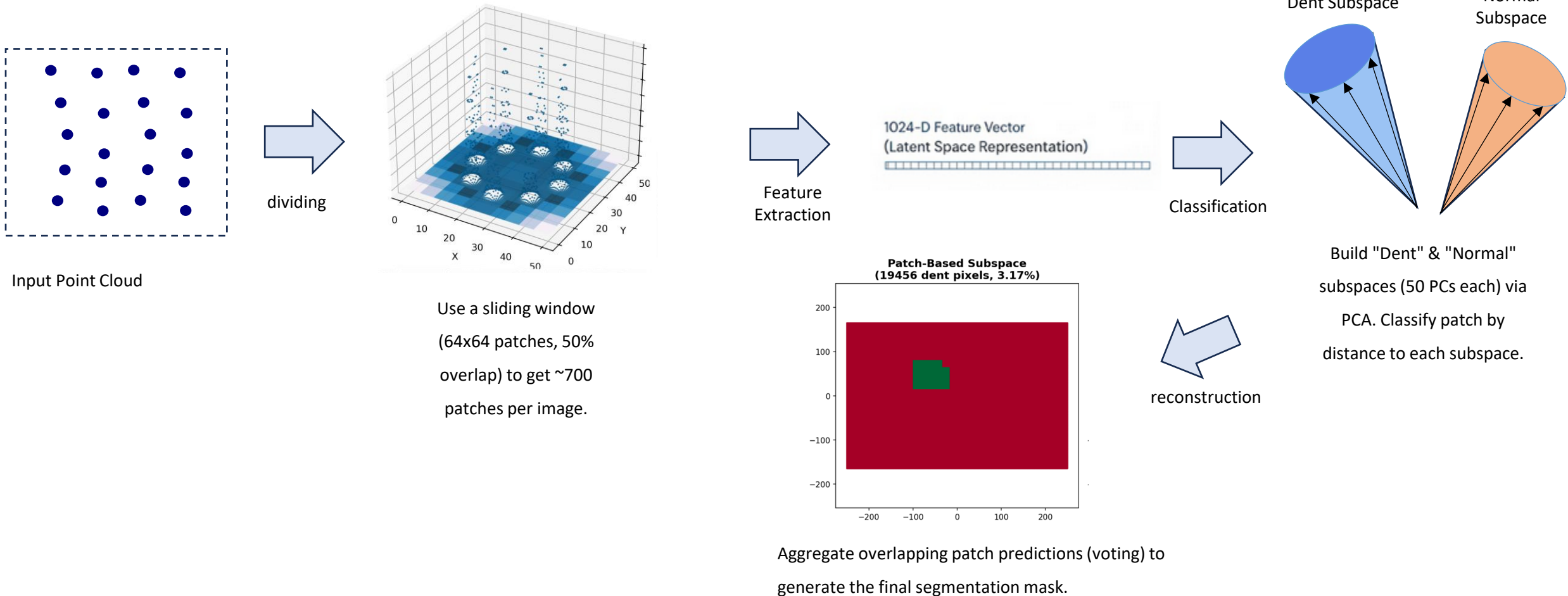
On a synthetic test sample, DentNet achieves **98.97% IoU**.

The output shows:

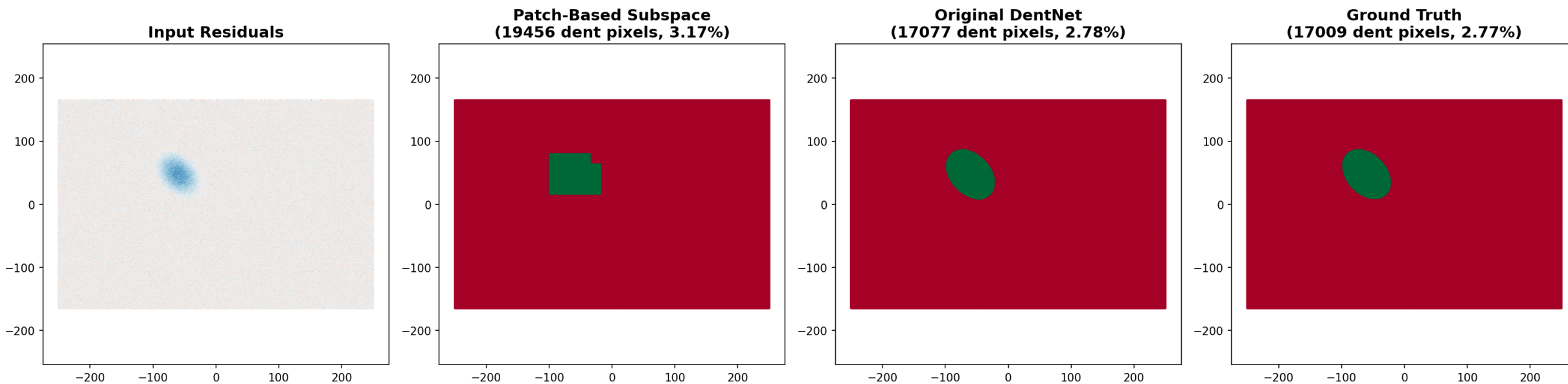
- **Input Residuals:** The dent is clearly visible as a blue depression.
- **Probability Map:** Hot colors indicate high confidence of a dent.
- **Final Prediction:** The boundaries are smooth, accurate, and (green) match the ground truth.



Our Approach: Patch-Based Subspace



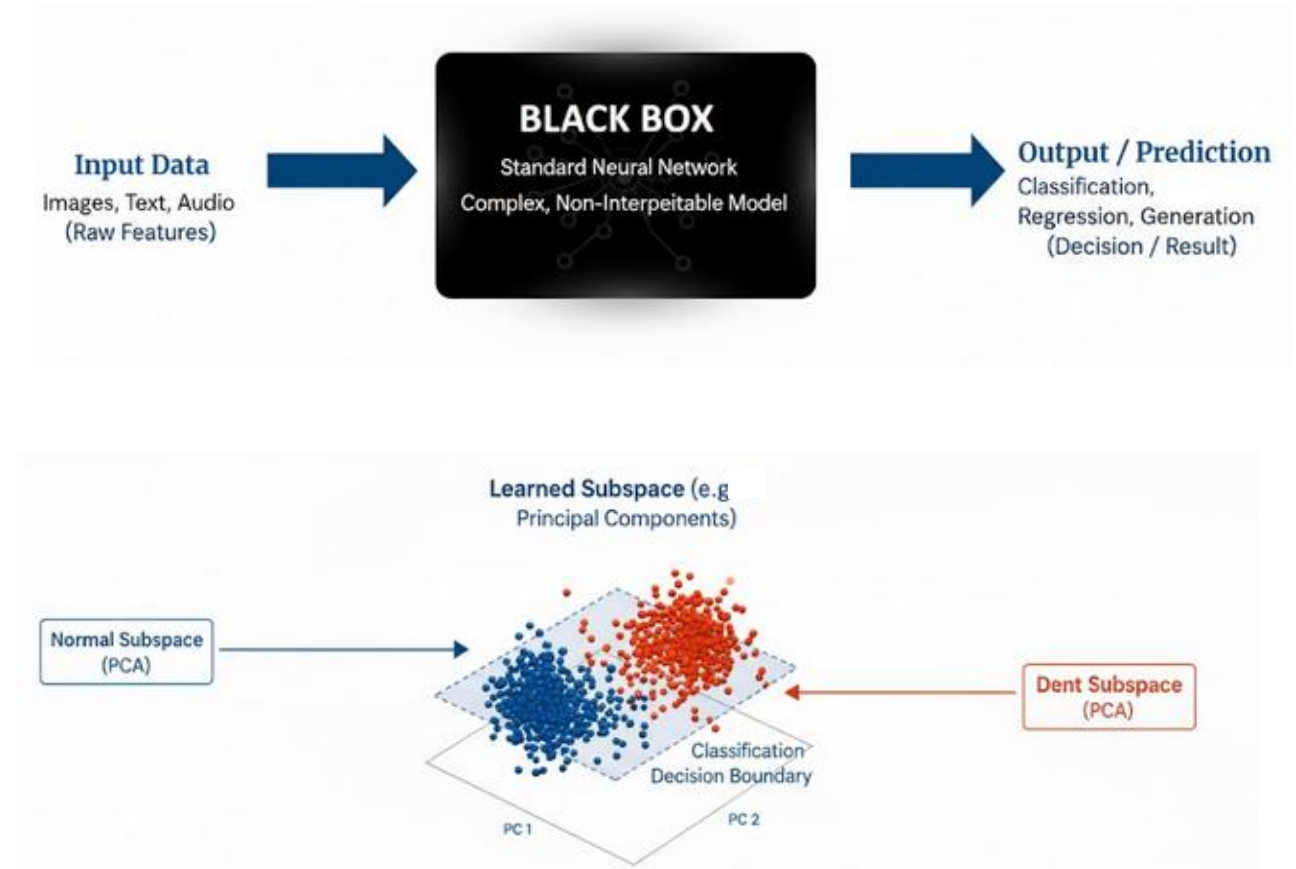
Results



Why Subspace Methods? Interpretability

Beyond the "Black Box"

- Unlike a standard neural network, the subspace approach is highly interpretable.
 - We can visualize the learned patterns.
 - We can see which features define "dent-ness".
 - We can quantitatively explain *why* a patch was classified (its distance to the subspaces).
 - This allows for validation with domain experts.



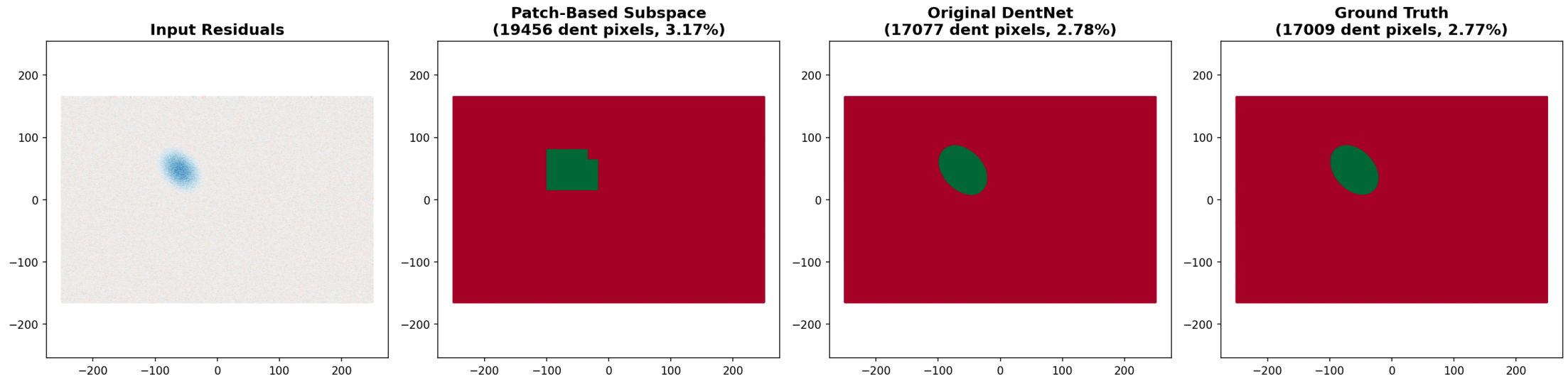
Comparison on Synthetic Sample

- **DentNet**

- **98.97% IoU**
- Very accurate, smooth, circular boundaries.
- Very close to ground truth.

Patch-Based Subspace

- **74.92% IoU**
- Lower performance, blocky boundaries (due to 64x64 patches).
- Slightly overshoots the dent area.
- However, the classification is fully interpretable.



Real-World Challenges

Struggles on Real Data

- When tested on real aircraft scans, both methods struggled.

Sample A (0.78mm dent):

- **DentNet:** 41.61% False Positives. Detects scan "strip boundaries" as dents.
- **Patch-Based:** 20.65% False Positives. Better, but still problematic.

Analysis: Why Do Both Methods Fail?

Domain Gap

Training data (clean, synthetic) lacks real-world scan artifacts like stitching "strip boundaries". The models misinterpret these as anomalies.

Quadric Fitting Issue

Training: Fit **excludes** dent points (uses ground truth). Inference: Fit **includes** all points (no mask), "flattening" the dent and making it invisible.

Magnitude Mismatch

Training was on 1-3mm dents. Real data was 0.78mm (too shallow) or 15mm (too deep), falling outside the trained range.

Summary

- Implemented **DentNet** (99.24% synthetic IoU) and a **Patch-Based Subspace** approach.
- **Synthetic:** DentNet is highly accurate (98.97%); Patch-based is interpretable but less precise (74.92%).
- **Real Data:** Both methods struggle. The patch-based approach shows slightly better generalization to noise (20.65% False Positive vs 41.61% False Positive).
- **Connection:** This confirms our initial finding: patch-based/sliding-window methods can be more robust to noise and domain gaps.
- **Main Lesson:** The domain gap and preprocessing (quadric fitting) are the most significant challenges to solve for real-world deployment.

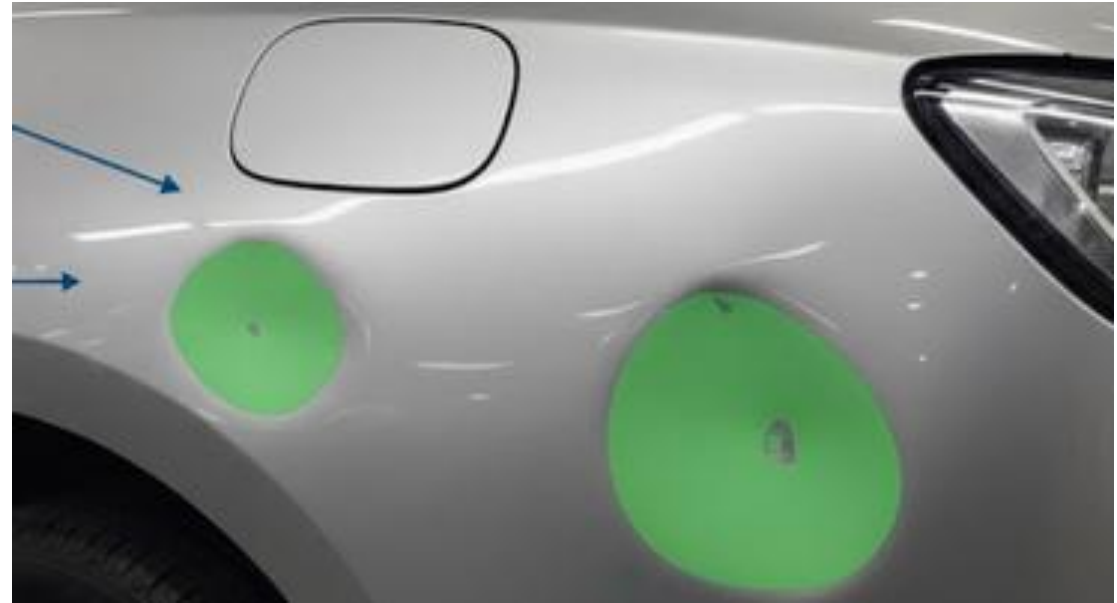
Future Work

Test on Real World Samples

- Fine-tune the model on a small, curated set of labeled real-world scans to bridge the domain gap.

Robust Quadric Fitting

- Use RANSAC to automatically exclude outliers (like dents) from the initial surface fit.



References

- [1] Kazuhiro Fukui. “Subspace Methods”. In: Jan. 2020, pp. 1–5. ISBN: 978-3-030-03243-2. DOI: 10.1007/978-3-030-03243-2_708-1.
- [2] Kazuhiro Fukui and Atsuto Maki. “Difference Subspace and Its Generalization for Subspace-Based Methods”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 37.11 (2015), pp. 2164–2177. DOI: 10.1109/TPAMI.2015.2408358.
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- [4] Yosuke Igarashi and Kazuhiro Fukui. “3D Object Recognition Based on Canonical Angles between Shape Subspaces”. In: *Computer Vision – ACCV 2010*. Ed. by Ron Kimmel, Reinhard Klette, and Akihiro Sugimoto. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 580–591. ISBN: 978-3-642-19282-1.
- [5] “Point Cloud Processing and Compression”. In: *IEEE Transactions on Circuits and Systems for Video Technology* 30.10 (2020), pp. 3856–3856. DOI: 10.1109/TCSVT.2020.3019906.
- [6] Takao Yoshinuma, Hideitsu Hino, and Kazuhiro Fukui. “Personal Authentication Based on 3D Configuration of Micro-feature Points on Facial Surface”. In: Feb. 2016, pp. 433–446. ISBN: 978-3-319-29450-6. DOI: 10.1007/978-3-319-29451-3_35.